

The Beveridge Response to Destruction and Separation Shocks under Infinitely Elastic Vacancy Creation

Notes for review (not for the draft), October 7, 2026

Numbers are produced by `Programs baseline/gs_free_entry_beveridge.jl` (output `gs_free_entry_bev` block “Exposition tables”). The program checks every closed form below by assertion against a simulated impulse response of the same linear system. Equation numbers “GS (*n*)” refer to Gabrovski and Silva (2025), *Journal of Economic Dynamics and Control* 181, 105205.

1. Setting

We study the Gabrovski–Silva (GS) economy with one change: the destruction rate δ_t and the separation probability s_t are time-varying shocks. In the draft this is the AGS economy of `prop:ags` with $\sigma = \kappa = 0$, which `rem:nesting` identifies as GS extended with aggregate shocks to δ_t and s_t . Its labor-market block closes on its own, as under the conditions of `prop:independence`. The discount rate is r , so the discount factor is $1/(1+r)$. Output per match, y (recruiter compensation z/μ in our model), is constant. Exit is exogenous, so a business survives from t to $t+1$ with probability $1 - \delta_t$. Every vacated position is reposted at no cost, and there are no flow recruiting costs, so all recruiting costs are sunk.

Shock processes.

$$\delta_t = \bar{\delta} \exp(\tilde{\delta}_t), \quad s_t = \bar{s} \exp(\tilde{s}_t),$$

where $\tilde{\delta}_t$ and $\tilde{s}_t$ are mean-zero AR(1) processes and are the log deviations of δ_t and s_t . We study one shock at a time and write $\rho \in [0, 1)$ for the persistence of the shock under study, $\tilde{x}_t = \rho \tilde{x}_{t-1} + \varepsilon_t$ for $x \in \{\delta, s\}$, with ε_t mean zero. The total separation rate

$$\tau_t \equiv 1 - (1 - \delta_t)(1 - s_t)$$

moves with both. Its steady state is $\bar{\tau} = 1 - (1 - \bar{\delta})(1 - \bar{s})$.

Timing. We follow GS (their Fig. 2) and the draft (S10). δ_t and s_t are known at the start of t . Entry, production and matching take place within t . Separation and destruction take place at the end of t , at rates s_t and δ_t , and a match formed at t is exposed to destruction at the end of t but not to separation. So u_t and the vacancies inherited from $t - 1$ are fixed at t , while tightness θ_t and entry e_t respond at t .

Matching. $q(\theta) = A\theta^{-\eta_L}$ and $f(\theta) = \theta q(\theta)$, with $\eta_L \in (0, 1)$. GS use the den Haan–Ramey–Watson function; we use Cobb–Douglas, as in our calibration.

2. The nonlinear system

2.1. Vacancy value and flow value

A business opportunity costs x , drawn from $G(x) = (x/x_m)^\xi$ on $[0, x_m]$, and is taken if and only if $x \leq Q_t$. So $e_t = G(Q_t)$ and $Q_t = e_t^{1/\xi} x_m$. Under infinitely elastic creation ($\xi \rightarrow \infty$),

$$Q_t = x_m \quad \text{for all } t. \quad (1)$$

The flow value of a vacancy is defined as in GS (2), with the survival factor dated t :

$$K_t \equiv Q_t - \frac{1 - \delta_t}{1 + r} \mathbb{E}_t Q_{t+1} = x_m \frac{r + \delta_t}{1 + r}. \quad (2)$$

Whereas Q is constant, K_t is a known function of the exogenous δ_t . In GS δ is constant, so free entry fixes K as well (GS (13)). Here δ_t is the depreciation rate of the vacancy asset, so its flow value moves with δ_t . Holding K fixed would give a flow-cost DMP model (Remark 2). We substitute (2) wherever K appears.

2.2. Wage and job creation

The Nash wage is GS (7):

$$w_t = \phi[y - K_t + \theta_t K_t] + (1 - \phi)b. \quad (3)$$

The job creation condition is GS (10), with the survival factor dated t :

$$\frac{K_t}{q(\theta_t)} = \frac{1 - \delta_t}{1 + r} \mathbb{E}_t \left[y - w_{t+1} - K_{t+1} + (1 - s_{t+1}) \frac{K_{t+1}}{q(\theta_{t+1})} \right]. \quad (4)$$

The left side is the expected cost of a hire: the flow value of a vacancy times the expected time to fill. The right side is the expected profit from a match, which first produces at $t + 1$, and only

if the business survives destruction at the end of t . Profit has two parts: the flow profit net of the forgone vacancy value, $y - w_{t+1} - K_{t+1}$, and the hiring cost saved if the match continues, $(1 - s_{t+1})K_{t+1}/q(\theta_{t+1})$. A match formed at t first faces separation at the end of $t + 1$, so the relevant rate is s_{t+1} . The wage (3) does not depend on s : the separation terms in the firm's and the worker's surplus cancel in the Nash sharing rule.

Using the Nash wage (3) dated $t + 1$,

$$y - w_{t+1} - K_{t+1} = (1 - \phi)(y - b) - K_{t+1}[(1 - \phi) + \phi\theta_{t+1}].$$

So the term in brackets in (4) is

$$(1 - \phi)(y - b) + K_{t+1}\left[\frac{1 - s_{t+1}}{q(\theta_{t+1})} - \phi\theta_{t+1} - (1 - \phi)\right].$$

Multiply (4) by $1 + r$ and substitute (2) at t and $t + 1$. The job creation condition becomes an equation in θ and the two shocks alone:

$$\frac{x_m(r + \delta_t)}{q(\theta_t)} = (1 - \delta_t) \mathbb{E}_t \left\{ (1 - \phi)(y - b) + \frac{x_m(r + \delta_{t+1})}{1 + r} \left[\frac{1 - s_{t+1}}{q(\theta_{t+1})} - \phi\theta_{t+1} - (1 - \phi) \right] \right\}. \quad (5)$$

The left-hand side represents the average hiring cost. The right-hand side represents the benefit of a filled job. Its first term, $(1 - \phi)(y - b)$, is the firm's share of the match's flow surplus. In the second term, $x_m(r + \delta_{t+1})/(1 + r) = K_{t+1}$ is next period's flow value of a vacancy. It is multiplied by $(1 - s_{t+1})/q(\theta_{t+1})$, the hiring cost saved if the match continues; by $-\phi\theta_{t+1}$, the worker's share of that saving, $\phi fK/q$; and by $-(1 - \phi)$, the firm's share of the forgone flow value of the vacancy.

2.3. Laws of motion

Vacancies follow GS (8). Unemployment follows GS (9), with δ and τ dated $t - 1$ because the destruction they cause takes place at the end of $t - 1$:

$$v_t = (1 - \delta_{t-1})[1 - q(\theta_{t-1})]v_{t-1} + (1 - \delta_{t-1})s_{t-1}(1 - u_{t-1}) + e_t, \quad (6)$$

$$u_t = [1 - (1 - \delta_{t-1})f(\theta_{t-1})]u_{t-1} + \tau_{t-1}(1 - u_{t-1}), \quad (7)$$

$$\theta_t = v_t/u_t. \quad (8)$$

The vacancy pool (6) has three parts: unmatched vacancies from $t - 1$ that avoided destruction, vacancies reposted after a separation at a surviving business, and new entrants e_t . The first two are fixed before t . Call their sum the predetermined stock:

$$v_{pre,t} \equiv (1 - \delta_{t-1})\left\{ [1 - q(\theta_{t-1})]v_{t-1} + s_{t-1}(1 - u_{t-1}) \right\}, \quad v_t = v_{pre,t} + e_t.$$

Every variable on its right side is dated $t - 1$, which is what makes it predetermined at t . Unemployment (7) has two parts: last period's unemployed who were not hired, or were hired at a business that was then destroyed, and last period's employed who separated, at rate τ_{t-1} .

How free entry orders the system. Equation (5) contains only θ and the exogenous shocks, so it determines the path of θ on its own. Equation (7) then gives u , (8) gives $v = \theta u$, and (6) determines entry, $e_t = v_t - v_{pre,t}$. With finitely elastic creation the causation reverses: entry is set by Q , (6) determines v , and (4) determines Q .

3. Steady state

Following GS's Definition 2, with $Q = x_m$, the steady state satisfies

$$\frac{\bar{K}}{q(\bar{\theta})} = \frac{1 - \bar{\delta}}{r + \bar{\tau}}(y - \bar{w} - \bar{K}), \quad (9)$$

$$\bar{K} = x_m \frac{r + \bar{\delta}}{1 + r}, \quad (10)$$

$$\bar{w} = \phi[y - \bar{K} + \bar{\theta}\bar{K}] + (1 - \phi)b, \quad (11)$$

$$\bar{u} = \frac{\bar{\tau}}{\bar{\tau} + (1 - \bar{\delta})f(\bar{\theta})}, \quad (12)$$

$$\bar{e} = \bar{\delta}(\bar{v} + 1 - \bar{u}), \quad (13)$$

$$\bar{\theta} = \bar{v}/\bar{u}.$$

These are GS (12)–(16).

Equation (9) is the steady state of (4),

$$\frac{\bar{K}}{\bar{q}} = \frac{1 - \bar{\delta}}{1 + r} \left[y - \bar{w} - \bar{K} + (1 - \bar{s}) \frac{\bar{K}}{\bar{q}} \right],$$

multiplied by $1 + r$ and rearranged using $(1 + r) - (1 - \bar{\delta})(1 - \bar{s}) = r + \bar{\tau}$. Equation (10) is a user cost: the asset price x_m times the discount rate plus the destruction rate. The share of the user cost due to destruction is $\bar{\delta}/(r + \bar{\delta})$. Equation (12) can be written $(1 - \bar{\delta})\bar{f}\bar{u} = \bar{\tau}(1 - \bar{u})$: hires equal separations. Since $\bar{f}\bar{u} = \bar{q}\bar{v}$, this is also $(1 - \bar{\delta})\bar{q}\bar{v} = \bar{\tau}(1 - \bar{u})$.

For (13), the steady state of (6) gives

$$\bar{e} = \bar{v} - (1 - \bar{\delta})(1 - \bar{q})\bar{v} - (1 - \bar{\delta})\bar{s}(1 - \bar{u}) = \bar{\delta}\bar{v} + (1 - \bar{\delta})\bar{q}\bar{v} - (1 - \bar{\delta})\bar{s}(1 - \bar{u}).$$

Using hires equal separations, this is $\bar{\delta}\bar{v} + [\bar{\tau} - (1 - \bar{\delta})\bar{s}](1 - \bar{u})$, and $\bar{\tau} - (1 - \bar{\delta})\bar{s} = \bar{\delta}$. So entry replaces the positions destroyed, both vacant ($\bar{v}$) and filled ($1 - \bar{u}$). Positions vacated by

separations are reposted and need no replacement. The shares of entrants and of predetermined vacancies in the stock are therefore

$$\frac{\bar{e}}{\bar{v}} = \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}}, \quad \frac{\bar{v}_{pre}}{\bar{v}} = 1 - \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}}.$$

Filled positions far outnumber vacancies, $(1 - \bar{u})/\bar{v} \approx 26$ at our targets, so the entrant share is small when destruction is rare. Hires equal separations gives $(1 - \bar{u})/\bar{v} = (1 - \bar{\delta})\bar{q}/\bar{\tau}$: filled jobs per vacancy equal hires per vacancy times the expected duration of a job. An equivalent form of the entrant share is therefore

$$\frac{\bar{e}}{\bar{v}} = \frac{\bar{\delta}}{\bar{\tau}} [\bar{\tau} + (1 - \bar{\delta})\bar{q}].$$

Our calibration fixes $\bar{\tau}$ and the gross vacancy-filling rate $(1 - \bar{\delta})\bar{q}$, so the bracket is the same, 0.831, in every economy we compare. The entrant share is then proportional to the destruction share of separations, $\bar{\delta}/\bar{\tau}$.

4. Log-linear system

Hats denote log deviations from the steady state, $\hat{x}_t \equiv \ln x_t - \ln \bar{x}$. The shocks are already in logs, $\hat{\delta}_t = \tilde{\delta}_t$ and $\hat{s}_t = \tilde{s}_t$, so $\delta_t - \bar{\delta} \approx \bar{\delta} \tilde{\delta}_t$ and $s_t - \bar{s} \approx \bar{s} \tilde{s}_t$. The total separation rate moves by $\tau_t - \bar{\tau} = (1 - \bar{s})(\delta_t - \bar{\delta}) + (1 - \bar{\delta})(s_t - \bar{s})$, so

$$\hat{\tau}_t = (1 - \bar{s}) \frac{\bar{\delta}}{\bar{\tau}} \tilde{\delta}_t + \left(1 - \frac{\bar{\delta}}{\bar{\tau}}\right) \tilde{s}_t,$$

using $(1 - \bar{\delta})\bar{s} = \bar{\tau} - \bar{\delta}$. A one-percent rise in δ raises τ by $(1 - \bar{s})\bar{\delta}/\bar{\tau}$ percent, the destruction share of separations up to the factor $1 - \bar{s}$. A one-percent rise in s raises it by $1 - \bar{\delta}/\bar{\tau}$ percent, the separation share.

4.1. Job creation

Take logs of both sides of (5). On the left,

$$\ln \frac{x_m(r + \delta_t)}{q(\theta_t)} = \ln x_m + \ln(r + \bar{\delta}e^{\tilde{\delta}_t}) - \ln A + \eta_L \ln \theta_t,$$

whose first-order deviation is

$$\frac{\bar{\delta}}{r + \bar{\delta}} \tilde{\delta}_t + \eta_L \hat{\theta}_t.$$

On the right, the log of $1 - \delta_t$ deviates by $-[\bar{\delta}/(1 - \bar{\delta})]\tilde{\delta}_t$. Write Σ_{t+1} for the term in braces in (5). In the steady state of (5), $x_m(r + \bar{\delta})/\bar{q} = (1 - \bar{\delta})\bar{\Sigma}$. Only the second term of Σ_{t+1} varies,

and its deviation is

$$d\Sigma_{t+1} = \frac{x_m}{1+r} \left\{ \bar{\delta} \tilde{\delta}_{t+1} \left[\frac{1-\bar{s}}{\bar{q}} - \phi \bar{\theta} - (1-\phi) \right] + (r+\bar{\delta}) \left[(1-\bar{s}) \frac{\eta_L}{\bar{q}} - \phi \bar{\theta} \right] \hat{\theta}_{t+1} - (r+\bar{\delta}) \frac{\bar{s}}{\bar{q}} \tilde{s}_{t+1} \right\},$$

using $d(1/q) = (\eta_L/\bar{q})\hat{\theta}$, $d\theta = \bar{\theta}\hat{\theta}$ and $ds = \bar{s}\tilde{s}$. Divide by $\bar{\Sigma} = x_m(r+\bar{\delta})/[(1-\bar{\delta})\bar{q}]$, and use $\bar{\theta}\bar{q} = \bar{f}$:

$$\hat{\Sigma}_{t+1} = \frac{1-\bar{\delta}}{1+r} \left\{ [1-\bar{s}-\phi\bar{f}-(1-\phi)\bar{q}] \frac{\bar{\delta}}{r+\bar{\delta}} \tilde{\delta}_{t+1} + [(1-\bar{s})\eta_L - \phi\bar{f}] \hat{\theta}_{t+1} - \bar{s} \tilde{s}_{t+1} \right\}.$$

Distribute $1-\bar{\delta}$ inside the braces. Since $(1-\bar{\delta})(1-\bar{s}) = 1-\bar{\tau}$,

$$\hat{\Sigma}_{t+1} = \frac{1}{1+r} \left\{ [1-\bar{\tau}-\phi(1-\bar{\delta})\bar{f}-(1-\phi)(1-\bar{\delta})\bar{q}] \frac{\bar{\delta}}{r+\bar{\delta}} \tilde{\delta}_{t+1} + [(1-\bar{\tau})\eta_L - \phi(1-\bar{\delta})\bar{f}] \hat{\theta}_{t+1} - (1-\bar{\delta})\bar{s} \tilde{s}_{t+1} \right\}.$$

The matching rates now appear as $(1-\bar{\delta})\bar{f}$ and $(1-\bar{\delta})\bar{q}$, the effective job-finding and vacancy-filling rates: the probabilities that a match forms and its business survives to production. These are the rates we calibrate, 0.41 and 0.80. Setting the left-side deviation equal to $-[\bar{\delta}/(1-\bar{\delta})]\tilde{\delta}_t + \mathbb{E}_t \hat{\Sigma}_{t+1}$, moving the $\tilde{\delta}_t$ term to the right and collecting terms,

$$\boxed{\eta_L \hat{\theta}_t = - \left[\frac{\bar{\delta}}{r+\bar{\delta}} + \frac{\bar{\delta}}{1-\bar{\delta}} \right] \tilde{\delta}_t + \frac{1}{1+r} \mathbb{E}_t \left\{ [1-\bar{\tau}-\phi(1-\bar{\delta})\bar{f}-(1-\phi)(1-\bar{\delta})\bar{q}] \frac{\bar{\delta}}{r+\bar{\delta}} \tilde{\delta}_{t+1} + [(1-\bar{\tau})\eta_L - \phi(1-\bar{\delta})\bar{f}] \hat{\theta}_{t+1} - (1-\bar{\delta})\bar{s} \tilde{s}_{t+1} \right\}} \quad (14)$$

Only $\hat{\theta}$ and the shocks appear. Next period's terms are discounted at $1/(1+r)$, and survival enters through the probability that the match continues, $1-\bar{\tau}$, and through the effective matching rates.

The weights in (14) read as follows. The coefficient η_L on $\hat{\theta}_t$ says that a tighter market lengthens the time to fill, so a hire costs more. The weight $\bar{\delta}/(r+\bar{\delta})$ on $\tilde{\delta}_t$ is the flow-value effect: a one-percent rise in δ raises the flow value $K_t = x_m(r+\delta_t)/(1+r)$ by $\bar{\delta}/(r+\bar{\delta})$ percent, the share of the user cost due to destruction, and so raises the cost of a hire. This share is close to one when $\bar{\delta} \gg r$ and close to $\bar{\delta}/r$ when $\bar{\delta} \ll r$. The weight $\bar{\delta}/(1-\bar{\delta})$ is the survival effect: a hire pays off only if the business survives, so destruction lowers the value of a hire. It is about $\bar{\delta}$, which is small.

The terms in $\tilde{\delta}_{t+1}$ are nonzero only if the shock persists. A higher flow value next period raises the hiring cost saved by a match that continues, with probability $1-\bar{\tau}$. Part of that saving goes to the worker through her outside option, $-\phi(1-\bar{\delta})\bar{f}$. A higher flow value also lowers the firm's flow profit, since a matched firm forgoes it, $-(1-\phi)(1-\bar{\delta})\bar{q}$. In the same way,

a tighter market next period raises the hiring cost a continuing match saves, $(1 - \bar{\tau})\eta_L$, but it also raises the worker's outside option and hence the wage, $-\phi(1 - \bar{\delta})\bar{f}$.

The separation shock enters only through $-(1 - \bar{\delta})\bar{s}\tilde{s}_{t+1}$. A higher separation rate next period lowers the probability that a match continues, and with it the hiring cost the match saves. Unlike δ , s does not move the flow value of a vacancy, because K_t depends only on δ_t . And because the term is dated $t + 1$, an iid separation shock does not move tightness at all.

4.2. Unemployment

Write the right side of (7) as $U(u_{t-1}, \delta_{t-1}, \tau_{t-1}, \theta_{t-1}) = [1 - (1 - \delta_{t-1})A\theta_{t-1}^{1-\eta_L}]u_{t-1} + \tau_{t-1}(1 - u_{t-1})$. Its partial derivatives at the steady state are

$$\begin{aligned}\frac{\partial U}{\partial u_{t-1}} &= 1 - (1 - \bar{\delta})\bar{f} - \bar{\tau}, & \frac{\partial U}{\partial \delta_{t-1}} &= \bar{f}\bar{u}, \\ \frac{\partial U}{\partial \tau_{t-1}} &= 1 - \bar{u}, & \frac{\partial U}{\partial \theta_{t-1}} &= -(1 - \bar{\delta})\bar{u}(1 - \eta_L)\frac{\bar{f}}{\bar{\theta}}.\end{aligned}$$

Hence, with $\delta_{t-1} - \bar{\delta} = \bar{\delta}\tilde{\delta}_{t-1}$ and $\tau_{t-1} - \bar{\tau} = \bar{\tau}\hat{\tau}_{t-1}$,

$$u_t - \bar{u} = [1 - (1 - \bar{\delta})\bar{f} - \bar{\tau}](u_{t-1} - \bar{u}) + \bar{f}\bar{u}\bar{\delta}\tilde{\delta}_{t-1} + (1 - \bar{u})\bar{\tau}\hat{\tau}_{t-1} - (1 - \bar{\delta})\bar{u}(1 - \eta_L)\bar{f}\hat{\theta}_{t-1}.$$

Divide by $\bar{u}$, and use $(1 - \bar{u})\bar{\tau}/\bar{u} = (1 - \bar{\delta})\bar{f}$ from (12). Define

$$\lambda \equiv 1 - (1 - \bar{\delta})\bar{f} - \bar{\tau}, \quad \omega \equiv (1 - \bar{\delta})\bar{f}(1 - \eta_L).$$

Then

$$\hat{u}_t = \lambda \hat{u}_{t-1} + \bar{f}\bar{\delta}\tilde{\delta}_{t-1} + (1 - \bar{\delta})\bar{f}\hat{\tau}_{t-1} - \omega \hat{\theta}_{t-1}. \quad (15)$$

Substitute $\hat{\tau}_{t-1}$. The two coefficients on $\tilde{\delta}_{t-1}$ combine as

$$\bar{f}\bar{\delta} + (1 - \bar{\delta})(1 - \bar{s})\bar{f}\frac{\bar{\delta}}{\bar{\tau}} = \bar{f}\frac{\bar{\delta}}{\bar{\tau}}[\bar{\tau} + (1 - \bar{\tau})] = \bar{f}\frac{\bar{\delta}}{\bar{\tau}},$$

because $(1 - \bar{\delta})(1 - \bar{s}) = 1 - \bar{\tau}$, and the coefficient on $\tilde{s}_{t-1}$ is $(1 - \bar{\delta})\bar{f}(1 - \bar{\delta}/\bar{\tau})$. So

$$\boxed{\hat{u}_t = \lambda \hat{u}_{t-1} + \bar{f}\frac{\bar{\delta}}{\bar{\tau}}\tilde{\delta}_{t-1} + (1 - \bar{\delta})\bar{f}\left(1 - \frac{\bar{\delta}}{\bar{\tau}}\right)\tilde{s}_{t-1} - \omega \hat{\theta}_{t-1}} \quad (16)$$

The coefficient λ is persistence. An extra unemployed worker is still unemployed next month unless hired at a surviving business, with probability $1 - (1 - \bar{\delta})\bar{f}$, and the $\bar{\tau}$ term reflects that one fewer employed worker means one fewer separation. The destruction weight has two channels, visible in (15). The first, $\bar{f}\bar{\delta}$, is lost matches: new matches at a destroyed business never

produce. The second is separations. In steady state the inflow into unemployment equals the outflow, $(1 - \bar{\delta})\bar{f}\bar{u}$, so a one-percent rise in separations raises unemployment by $(1 - \bar{\delta})\bar{f}$ percent, and separations rise by $(1 - \bar{s})\bar{\delta}/\bar{\tau}$ percent per percent of δ . Together the two channels give $\bar{f}\bar{\delta}/\bar{\tau}$, and the second dominates. The unemployment loading is therefore proportional to the destruction share of separations, $\bar{\delta}/\bar{\tau}$. The separation weight, $(1 - \bar{\delta})\bar{f}(1 - \bar{\delta}/\bar{\tau})$, has only the second channel, since s does not hit matches before they produce. It is the effective job-finding rate times the separation share of separations. Finally, ω is the hiring channel: lower tightness at $t - 1$ means fewer hires, and those workers are still unemployed at t .

4.3. Vacancies

Take the deviation of $v_t = v_{pre,t} + e_t$ and divide by $\bar{v}$:

$$\hat{v}_t = \frac{\bar{v}_{pre}}{\bar{v}} \hat{v}_{pre,t} + \frac{\bar{e}}{\bar{v}} \hat{e}_t.$$

Substituting the shares implied by (13),

$$\hat{v}_t = \left[1 - \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}} \right] \hat{v}_{pre,t} + \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}} \hat{e}_t \quad (17)$$

For the predetermined stock, the partial derivatives of $v_{pre,t} = (1 - \delta_{t-1})\{[1 - A\theta_{t-1}^{-\eta_L}]v_{t-1} + s_{t-1}(1 - u_{t-1})\}$ at the steady state are

$$\begin{aligned} \frac{\partial v_{pre,t}}{\partial \delta_{t-1}} &= -[(1 - \bar{q})\bar{v} + \bar{s}(1 - \bar{u})] = -\frac{\bar{v}_{pre}}{1 - \bar{\delta}}, & \frac{\partial v_{pre,t}}{\partial v_{t-1}} &= (1 - \bar{\delta})(1 - \bar{q}), \\ \frac{\partial v_{pre,t}}{\partial \theta_{t-1}} &= (1 - \bar{\delta})\eta_L \frac{\bar{q}}{\bar{\theta}} \bar{v}, & \frac{\partial v_{pre,t}}{\partial u_{t-1}} &= -(1 - \bar{\delta})\bar{s}, \\ \frac{\partial v_{pre,t}}{\partial s_{t-1}} &= (1 - \bar{\delta})(1 - \bar{u}). \end{aligned}$$

The first uses the steady state $\bar{v}_{pre} = (1 - \bar{\delta})[(1 - \bar{q})\bar{v} + \bar{s}(1 - \bar{u})]$, and the third uses $\partial q/\partial \theta = -\eta_L \bar{q}/\bar{\theta}$. Hence, with $\delta_{t-1} - \bar{\delta} = \bar{\delta}\tilde{\delta}_{t-1}$ and $s_{t-1} - \bar{s} = \bar{s}\tilde{s}_{t-1}$,

$$v_{pre,t} - \bar{v}_{pre} = -\frac{\bar{v}_{pre}}{1 - \bar{\delta}} \tilde{\delta}_{t-1} + (1 - \bar{\delta})[(1 - \bar{q})\bar{v} \hat{v}_{t-1} + \eta_L \bar{q}\bar{v} \hat{\theta}_{t-1} - \bar{s}\bar{u} \hat{u}_{t-1} + \bar{s}(1 - \bar{u}) \tilde{s}_{t-1}].$$

Dividing by $\bar{v}_{pre} = \bar{v} - \bar{e}$,

$$\hat{v}_{pre,t} = -\frac{\bar{\delta}}{1 - \bar{\delta}} \tilde{\delta}_{t-1} + \frac{1 - \bar{\delta}}{\bar{v} - \bar{e}} \left[(1 - \bar{q})\bar{v} \hat{v}_{t-1} + \eta_L \bar{q}\bar{v} \hat{\theta}_{t-1} - \bar{s}\bar{u} \hat{u}_{t-1} + \bar{s}(1 - \bar{u}) \tilde{s}_{t-1} \right] \quad (18)$$

Entry replaces the positions destroyed, so its weight in (17), $\bar{\delta}(\bar{v} + 1 - \bar{u})/\bar{v}$, is proportional to $\bar{\delta}$: 0.073 at D1 and 0.831 at $\bar{\delta} = \bar{\tau}$. The predetermined stock is the rest, 0.927 and 0.169.

In (18), a one-percent rise in δ_{t-1} destroys about $\bar{\delta}$ percent of the predetermined stock, while entrants posted at t are not exposed. The bracket collects the changes in the sources of the predetermined stock, in units of vacancies. More vacancies at $t - 1$ leave more unfilled ones $((1 - \bar{q})\bar{v} \hat{v}_{t-1})$. A tighter market fills fewer of them $(\eta_L \bar{q} \bar{v} \hat{\theta}_{t-1})$; after a destruction shock $\hat{\theta} < 0$, so vacancies fill faster and fewer carry over. And reposts come from the employed, who are fewer when u is higher $(-\bar{s}\bar{u} \hat{u}_{t-1})$, and they rise one for one with the separation rate $(\bar{s}(1 - \bar{u}) \hat{s}_{t-1})$. A separation shock thus adds the same positions to the vacancy pool that it adds workers to unemployment, which is the accounting behind Prop. 5, Part 1.

At free entry. With $\hat{v}_{t-1} = \hat{\theta}_{t-1} + \hat{u}_{t-1}$, the bracket in (18) becomes $(1 - \bar{q} + \eta_L \bar{q})\bar{v} \hat{\theta}_{t-1} + [(1 - \bar{q})\bar{v} - \bar{s}\bar{u}]\hat{u}_{t-1} + \bar{s}(1 - \bar{u}) \hat{s}_{t-1}$. Equation (17) then determines entry, not the vacancy stock:

$$\hat{e}_t = \frac{\bar{v}}{\bar{\delta}(\bar{v} + 1 - \bar{u})} \left\{ \hat{v}_t - \left[1 - \frac{\bar{\delta}(\bar{v} + 1 - \bar{u})}{\bar{v}} \right] \hat{v}_{pre,t} \right\}, \quad \hat{v}_t = \hat{\theta}_t + \hat{u}_t.$$

When the entrant weight is small, as at D1, a given gap between the target stock and the predetermined stock requires a large percentage change in entry.

4.4. Tightness

Taking logs of (8),

$$\boxed{\hat{v}_t = \hat{\theta}_t + \hat{u}_t} \tag{19}$$

4.5. Weights at the calibration

The numbers use our calibration, not GS's. It holds $\bar{f}$, $\bar{q}$ and $\bar{\tau}$ fixed and recalibrates s and ϕ at each $\bar{\delta}$. The discount rate is $r = 0.0033$ monthly.

Moving from $\bar{\delta} = \bar{\tau}$ to D1 cuts the unemployment loading by a factor of 12 but the flow-value loading only by a factor of 2. The first is proportional to $\bar{\delta}$. The second saturates, because $\bar{\delta}/(r + \bar{\delta})$ falls in proportion to $\bar{\delta}$ only once $\bar{\delta}$ is below r . The entrant share falls by the same factor of 11 as the destruction share of separations, since $\bar{e}/\bar{v} = (\bar{\delta}/\bar{\tau})[\bar{\tau} + (1 - \bar{\delta})\bar{q}]$ with a bracket fixed by our targets. So the unemployment loading and the entrant share both scale with $\bar{\delta}/\bar{\tau}$, while the flow-value loading does not. The other weights are essentially the same in both economies.

	D1 ($\bar{\delta} = 3.2\%/yr$)	CK-like ($\bar{\delta} = \bar{\tau}$)
$\bar{\delta}/(r + \bar{\delta})$ (flow value)	0.453	0.905
$\bar{f}\bar{\delta}/\bar{\tau}$ (unemployment)	0.036	0.423
$(1 - \bar{s})\bar{\delta}/\bar{\tau}$ (separations, δ shock)	0.085	1.000
$(1 - \bar{\delta})\bar{f}(1 - \bar{\delta}/\bar{\tau})$ (unemployment, s shock)	0.374	0
$(1 - \bar{\delta})\bar{f}$	0.410	0.410
λ	0.559	0.559
ω	0.164	0.164
$1 - \bar{\delta}(\bar{v} + 1 - \bar{u})/\bar{v}$ (predetermined)	0.927	0.169
$\bar{\delta}(\bar{v} + 1 - \bar{u})/\bar{v}$ (entrants)	0.073	0.831

Table 1: Weights of the log-linear system, per unit of $\tilde{\delta}$ or $\tilde{s}$.

5. Proposition

With λ and ω as defined in Section 4, let ρ be the persistence of the shock under study and define the common denominator

$$D \equiv \eta_L[1 + r - \rho(1 - \bar{\tau})] + \rho\phi(1 - \bar{\delta})\bar{f}$$

and the tightness coefficients

$$a \equiv - \frac{(1 + r)\frac{\bar{\delta}}{1 - \bar{\delta}} + \frac{\bar{\delta}}{r + \bar{\delta}}[1 + r - \rho(1 - \bar{\tau}) + \rho\phi(1 - \bar{\delta})\bar{f} + \rho(1 - \phi)(1 - \bar{\delta})\bar{q}]}{D}, \quad b \equiv - \frac{\rho(1 - \bar{\delta})\bar{s}}{D}. \quad (20)$$

Lemma 1 (Tightness under infinitely elastic vacancy creation). *Assume the setting of Section 1 with $\xi \rightarrow \infty$, and suppose $\phi(1 - \bar{\delta})\bar{f} < \eta_L(2 + r - \bar{\tau})$, which holds whenever $\phi \leq \eta_L$. Then the log-linear system (14)–(19) has a unique bounded solution, in which tightness depends only on the shocks:*

$$\hat{\theta}_t = a\tilde{\delta}_t + b\tilde{s}_t, \quad a < 0, \quad b \leq 0,$$

with a and b evaluated at the persistence of the respective shock, and $b = 0$ if and only if $\rho = 0$. The endogenous states u_t and $v_{pre,t}$ do not enter the policy rule for tightness.

Proof. A general linear policy rule would take the form $\hat{\theta}_t = a\tilde{\delta}_t + b\tilde{s}_t + b_u\hat{u}_t + b_v\hat{v}_{pre,t}$. Equation (14) contains neither u nor v_{pre} , because free entry pins Q and $K_t = x_m(r + \delta_t)/(1 + r)$ depends

only on the exogenous δ_t . Dividing it by η_L ,

$$\hat{\theta}_t = \Phi \mathbb{E}_t \hat{\theta}_{t+1} + c_0 \tilde{\delta}_t + c_1 \mathbb{E}_t \tilde{\delta}_{t+1} + c_2 \mathbb{E}_t \tilde{s}_{t+1}, \quad \Phi \equiv \frac{1}{1+r} \left[1 - \bar{\tau} - \frac{\phi(1-\bar{\delta})\bar{f}}{\eta_L} \right],$$

$$c_0 \equiv -\frac{1}{\eta_L} \left[\frac{\bar{\delta}}{r+\bar{\delta}} + \frac{\bar{\delta}}{1-\bar{\delta}} \right], \quad c_1 \equiv \frac{1 - \bar{\tau} - \phi(1-\bar{\delta})\bar{f} - (1-\phi)(1-\bar{\delta})\bar{q}}{\eta_L(1+r)} \frac{\bar{\delta}}{r+\bar{\delta}}, \quad c_2 \equiv -\frac{(1-\bar{\delta})\bar{s}}{\eta_L(1+r)}.$$

Iterating forward T periods,

$$\hat{\theta}_t = \sum_{k=0}^{T-1} \Phi^k \mathbb{E}_t (c_0 \tilde{\delta}_{t+k} + c_1 \tilde{\delta}_{t+k+1} + c_2 \tilde{s}_{t+k+1}) + \Phi^T \mathbb{E}_t \hat{\theta}_{t+T}.$$

Here $\Phi < (1 - \bar{\tau})/(1 + r) < 1$. And $\Phi > -1$ if and only if $1 - \bar{\tau} - \phi(1 - \bar{\delta})\bar{f}/\eta_L > -(1 + r)$, that is, $\phi(1 - \bar{\delta})\bar{f} < \eta_L(2 + r - \bar{\tau})$, which is the assumption. It holds whenever $\phi \leq \eta_L$, because $(1 - \bar{\delta})\bar{f} < 1 < 2 + r - \bar{\tau}$. For a bounded solution the last term vanishes as $T \rightarrow \infty$, so the unique bounded solution is the forward sum. It depends only on current and expected future shocks, so $b_u = b_v = 0$. With each shock an AR(1), every term of the sum is proportional to the current shock, so the rule is linear in $\tilde{\delta}_t$ and $\tilde{s}_t$.

To find a , set $\tilde{s} \equiv 0$, substitute $\hat{\theta}_t = a\tilde{\delta}_t$, $\mathbb{E}_t \hat{\theta}_{t+1} = a\rho\tilde{\delta}_t$ and $\mathbb{E}_t \tilde{\delta}_{t+1} = \rho\tilde{\delta}_t$ into (14), divide by $\tilde{\delta}_t$ and multiply by $1 + r$:

$$(1+r)\eta_L a = -(1+r) \left[\frac{\bar{\delta}}{r+\bar{\delta}} + \frac{\bar{\delta}}{1-\bar{\delta}} \right] + \rho \left[1 - \bar{\tau} - \phi(1-\bar{\delta})\bar{f} - (1-\phi)(1-\bar{\delta})\bar{q} \right] \frac{\bar{\delta}}{r+\bar{\delta}} + \rho \left[(1-\bar{\tau})\eta_L - \phi(1-\bar{\delta})\bar{f} \right] a.$$

Collecting the terms in a on the left gives aD on the left and the negative of the numerator of a on the right, which is (20). To find b , set $\tilde{\delta} \equiv 0$ and substitute $\hat{\theta}_t = b\tilde{s}_t$ and $\mathbb{E}_t \tilde{s}_{t+1} = \rho\tilde{s}_t$ in the same way:

$$(1+r)\eta_L b = -\rho(1-\bar{\delta})\bar{s} + \rho \left[(1-\bar{\tau})\eta_L - \phi(1-\bar{\delta})\bar{f} \right] b,$$

so $bD = -\rho(1-\bar{\delta})\bar{s}$. Since $\rho < 1$, $1+r-\rho(1-\bar{\tau}) > 0$, so $D > 0$ and every term of the numerator of a is positive. Hence $a < 0$, and $b \leq 0$ with equality if and only if $\rho = 0$. $\square$

Reading (20). The term common to the two denominators,

$$1 + r - \rho(1 - \bar{\tau}) = r + \bar{\tau} + (1 - \rho)(1 - \bar{\tau}),$$

is the rate at which a match discounts the shock's effects on its future payoffs: the discount rate, plus the separation rate, plus the rate at which the shock dies out. In D it multiplies η_L ,

the response of the expected cost of a hire to tightness. The second term of D , $\rho\phi(1 - \bar{\delta})\bar{f}$, is the wage feedback: lower tightness lowers the worker's effective job-finding rate and hence her outside option, which props up the firm's surplus. At $\rho = 1$, $D = \eta_L(r + \bar{\tau}) + \phi(1 - \bar{\delta})\bar{f}$, the same combination of $r + \bar{\tau}$ and $\phi(1 - \bar{\delta})\bar{f}$ that appears in the steady-state job creation condition once the wage is substituted.

The numerator of a has a survival part, $(1 + r)\bar{\delta}/(1 - \bar{\delta})$, which is small, and a flow-value part, $\bar{\delta}/(r + \bar{\delta})$, times a bracket. The bracket starts from the same effective rate, which already nets out that a persistently high flow value raises the hiring cost a continuing match saves, through $\rho(1 - \bar{\tau})$. Two persistence terms then add to it: part of that saving goes to the worker, $\rho\phi(1 - \bar{\delta})\bar{f}$, and a matched firm forgoes the higher flow value, $\rho(1 - \phi)(1 - \bar{\delta})\bar{q}$. At $\rho = 0$, $a = -[\bar{\delta}/(r + \bar{\delta}) + \bar{\delta}/(1 - \bar{\delta})]/\eta_L$.

The numerator of b is only $\rho(1 - \bar{\delta})\bar{s}$: a persistent rise in separations lowers the probability that next period's match continues. There is no flow-value term, because s does not depreciate the vacancy. This is the source of the asymmetry between the two shocks.

Remark 1 (Block recursivity). *Given Lemma 1, the states are solved backward. Equation (16) gives $\hat{u}$, and (19) gives $\hat{v} = \hat{\theta} + \hat{u}$. The predetermined stock $v_{pre,t} = (1 - \delta_{t-1})\{[1 - q(\theta_{t-1})]\theta_{t-1}u_{t-1} + s_{t-1}(1 - u_{t-1})\}$ depends only on $(u_{t-1}, \theta_{t-1}, \delta_{t-1}, s_{t-1})$ and has no own lag. It feeds into nothing except entry, $e_t = v_t - v_{pre,t}$. The system thus has one unstable root, $1/\Phi$, for one jump variable, and one stable root, λ , for one dynamic state, u , so the Blanchard–Kahn condition holds.*

*The structure breaks with finite ξ . Entry $e = G(Q)$ then links v_{pre} to Q , and Q enters the job creation condition. Tightness depends on the states, the policy rule must include state coefficients, and the closed form is lost. The same happens when the conditions of **prop:independence** fail, through N , C or χ^c .*

For each shock $x \in \{\delta, s\}$, write c_x for its tightness coefficient ($c_\delta = a$, $c_s = b$) and ℓ_x for its unemployment loading in (16),

$$\ell_\delta \equiv \bar{f} \frac{\bar{\delta}}{\bar{\tau}}, \quad \ell_s \equiv (1 - \bar{\delta})\bar{f} \left(1 - \frac{\bar{\delta}}{\bar{\tau}}\right),$$

and, when $c_x \neq 0$, define the ratio of the unemployment loading to the tightness loading,

$$R_x \equiv \frac{\ell_x}{|c_x|}.$$

In the CK case, $\bar{\delta} = \bar{\tau}$, so $R_\delta = \bar{f}/|a|$ and there is no separation margin ($\bar{s} = 0$).

Proposition 1 (Destruction and separation shocks under infinitely elastic vacancy creation). *Under the assumptions of Lemma 1, consider an innovation $\varepsilon_0 > 0$ to shock $x \in \{\delta, s\}$. Let $\lambda \equiv 1 - (1 - \bar{\delta})\bar{f} - \bar{\tau} \in [0, 1)$ be the persistence of unemployment, one minus the effective job-finding rate and the separation rate, and let $\omega \equiv (1 - \eta_L)(1 - \bar{\delta})\bar{f}$ be the effective job-finding rate times its elasticity to tightness. To first order,*

$$\hat{\theta}_t = c_x \rho^t \varepsilon_0, \quad \hat{u}_t = (\ell_x + \omega |c_x|) \varepsilon_0 \sum_{j=0}^{t-1} \lambda^{t-1-j} \rho^j, \quad \hat{v}_t = \hat{\theta}_t + \hat{u}_t,$$

so unemployment is unchanged on impact and rises thereafter. Moreover, when $c_x \neq 0$:

1. Vacancies fall at $h = 1$ if and only if

$$\rho > \omega + R_x. \quad (21)$$

Once vacancies rise above steady state, they stay above it.

2. The slope of $v_t - \bar{v}$ on $u_t - \bar{u}$ along the impulse response is

$$\bar{\theta} \left[1 - \frac{\rho(1 - \lambda^2)}{(1 + \rho\lambda)(\omega + R_x)} \right], \quad (22)$$

which is negative if and only if

$$\frac{\rho(1 - \lambda^2)}{1 + \rho\lambda} > \omega + R_x. \quad (23)$$

3. For the separation shock, $R_s = (1 - \bar{\delta})\bar{f} D/(\rho\bar{\tau})$. If

$$(1 - \bar{\delta})\bar{f} \left[\eta_L + \frac{\phi(1 - \bar{\delta})\bar{f}}{\bar{\tau}} \right] \geq 1, \quad (24)$$

then after a separation shock $\hat{v}_t > 0$ for every $t \geq 1$ and the slope is positive, for every $\rho \in (0, 1)$.

If $\rho = 0$, both conditions fail for the destruction shock, and the separation shock leaves tightness unchanged at every date. In both cases $v_t = \bar{\theta}u_t$ after impact and the slope equals $\bar{\theta} > 0$.

Both conditions compare the persistence of the shock with the same threshold, $\omega + R_x$. At $h = 1$, the tightness fall that survives one period, $|c_x|\rho$, must exceed the rise in unemployment. That rise has two parts: the hires lost at date 0, $\omega|c_x|$, and the direct effect of the shock, $\ell_x = R_x|c_x|$. So $\rho > \omega$ is necessary, and R_x is the extra persistence needed to outweigh the direct effect. Over the whole path, persistence is discounted by unemployment's own persistence,

since $\rho(1-\lambda^2)/(1+\rho\lambda) < \rho$, so (23) is the stricter condition. The two shocks differ only through R_x . Condition (24) says that job finding is fast relative to separation and that workers have some bargaining power, so that wage feedback cushions tightness.

Proof. The innovation sets $\tilde{x}_t = \rho^t \varepsilon_0$ and the other shock to zero, so by Lemma 1 $\hat{\theta}_t = c_x \rho^t \varepsilon_0$.

Step 1: unemployment. u_0 is predetermined, so $\hat{u}_0 = 0$. Substituting $\hat{\theta}_{t-1} = -|c_x| \rho^{t-1} \varepsilon_0$ and $\tilde{x}_{t-1} = \rho^{t-1} \varepsilon_0$ into (16),

$$\hat{u}_t = \lambda \hat{u}_{t-1} + \ell_x \rho^{t-1} \varepsilon_0 + \omega |c_x| \rho^{t-1} \varepsilon_0 = \lambda \hat{u}_{t-1} + (\ell_x + \omega |c_x|) \rho^{t-1} \varepsilon_0.$$

Iterating from $\hat{u}_0 = 0$ gives $\hat{u}_1 = (\ell_x + \omega |c_x|) \varepsilon_0$, $\hat{u}_2 = (\ell_x + \omega |c_x|)(\lambda + \rho) \varepsilon_0$, and in general $\hat{u}_t = (\ell_x + \omega |c_x|) \varepsilon_0 g_t$ with $g_t \equiv \sum_{j=0}^{t-1} \lambda^{t-1-j} \rho^j$. Since $\lambda, \rho \geq 0$ and $g_1 = 1$, $\hat{u}_t > 0$ for $t \geq 1$. By (19), $\hat{v}_t = \hat{\theta}_t + \hat{u}_t$. When $c_x \neq 0$, $\ell_x + \omega |c_x| = |c_x|(\omega + R_x)$.

Step 2: part 1. With $g_1 = 1$,

$$\hat{v}_1 = -|c_x| \rho \varepsilon_0 + |c_x|(\omega + R_x) \varepsilon_0 = |c_x|(\omega + R_x - \rho) \varepsilon_0,$$

which is negative if and only if (21) holds. For $t \geq 1$ and $\rho > 0$, divide $\hat{v}_t = |c_x|[-\rho^t + (\omega + R_x)g_t] \varepsilon_0$ by $|c_x| \rho^t \varepsilon_0 > 0$, and use $g_t/\rho^t = \rho^{-1} \sum_{i=0}^{t-1} (\lambda/\rho)^i$:

$$\hat{v}_t < 0 \iff \rho > (\omega + R_x) \sum_{i=0}^{t-1} \left(\frac{\lambda}{\rho}\right)^i. \quad (25)$$

The right side is strictly increasing in t . So once the inequality fails, it fails at all later dates.

Step 3: part 2. In levels, $u_t - \bar{u} = \bar{u} |c_x|(\omega + R_x) \varepsilon_0 g_t$ and $v_t - \bar{v} = \bar{v} |c_x|[-\rho^t + (\omega + R_x)g_t] \varepsilon_0$, with $g_0 = 0$. For $\rho \neq \lambda$, $g_t = (\rho^t - \lambda^t)/(\rho - \lambda)$; the case $\rho = \lambda$ follows by continuity. Summing geometric series,

$$\sum_{t \geq 0} \rho^t g_t = \frac{\rho}{(1-\rho^2)(1-\rho\lambda)}, \quad \sum_{t \geq 0} g_t^2 = \frac{1+\rho\lambda}{(1-\rho^2)(1-\lambda^2)(1-\rho\lambda)}.$$

The first uses $\frac{1}{1-\rho^2} - \frac{1}{1-\rho\lambda} = \frac{\rho(\rho-\lambda)}{(1-\rho^2)(1-\rho\lambda)}$. The second uses $(1-\rho\lambda)(1-\lambda^2) - 2(1-\rho^2)(1-\lambda^2) + (1-\rho^2)(1-\rho\lambda) = (\rho-\lambda)^2(1+\rho\lambda)$. Therefore

$$\frac{\sum_t (v_t - \bar{v})(u_t - \bar{u})}{\sum_t (u_t - \bar{u})^2} = \frac{\bar{v}}{\bar{u}} \left[1 - \frac{1}{\omega + R_x} \frac{\sum_t \rho^t g_t}{\sum_t g_t^2} \right] = \bar{\theta} \left[1 - \frac{\rho(1-\lambda^2)}{(1+\rho\lambda)(\omega + R_x)} \right],$$

which is (22). It is negative if and only if (23) holds.

Step 4: part 3. By Lemma 1, $|b| = \rho(1-\bar{\delta})\bar{s}/D$. Using $(1-\bar{\delta})\bar{s} = \bar{\tau} - \bar{\delta} = \bar{\tau}(1-\bar{\delta}/\bar{\tau})$,

$$R_s = \frac{(1-\bar{\delta})\bar{f}(1-\bar{\delta}/\bar{\tau})D}{\rho\bar{\tau}(1-\bar{\delta}/\bar{\tau})} = \frac{(1-\bar{\delta})\bar{f}D}{\rho\bar{\tau}}.$$

Since $(1+r)/\rho > 1+r$,

$$\frac{D}{\rho} = \eta_L \left[\frac{1+r}{\rho} - (1-\bar{\tau}) \right] + \phi(1-\bar{\delta})\bar{f} > \eta_L(r+\bar{\tau}) + \phi(1-\bar{\delta})\bar{f},$$

so

$$R_s > (1-\bar{\delta})\bar{f} \left[\eta_L \frac{r+\bar{\tau}}{\bar{\tau}} + \frac{\phi(1-\bar{\delta})\bar{f}}{\bar{\tau}} \right] > (1-\bar{\delta})\bar{f} \left[\eta_L + \frac{\phi(1-\bar{\delta})\bar{f}}{\bar{\tau}} \right] \geq 1$$

under (24). Then for $t \geq 1$, $(\omega + R_s)g_t > g_t \geq \rho^{t-1} \geq \rho^t$, because g_t contains the term ρ^{t-1} , so $\hat{v}_t = |b|[-\rho^t + (\omega + R_s)g_t]\varepsilon_0 > 0$. And $\rho(1-\lambda^2)/(1+\rho\lambda) < \rho < 1 < \omega + R_s$, so the slope (22) is positive.

Step 5: $\rho = 0$. For the destruction shock, the left sides of (21) and (23) are zero while $\omega + R_\delta > 0$, so both conditions fail. Since $\hat{\theta}_t = a \cdot 0^t \varepsilon_0 = 0$ for $t \geq 1$, $\hat{v}_t = \hat{u}_t$ and $v_t = \bar{\theta}u_t$ after impact, and the slope (22) is $\bar{\theta}$. For the separation shock, $b = 0$, so $\hat{\theta}_t = 0$ at every date and $\hat{v}_t = \hat{u}_t$ throughout. The slope is again $\bar{\theta}$. $\square$

Corollary 1 (Vacancies below steady state at every horizon). *Under the assumptions of Proposition 1, with $c_x \neq 0$, $\hat{v}_t < 0$ for every $t \geq 0$ if and only if $\rho - \lambda \geq \omega + R_x$. This requires $\rho > \lambda + \omega$, which is 0.72 at our calibration.*

Proof. In (25), if $\rho \leq \lambda$ the right side grows without bound, so vacancies eventually rise above steady state. If $\rho > \lambda$ it converges from below to $(\omega + R_x)\rho/(\rho - \lambda)$, so the inequality holds at every t if and only if $\rho - \lambda \geq \omega + R_x$. $\square$

6. Interpretation

Why tightness falls. At $\rho = 0$, (14) reads $\eta_L \hat{\theta}_0 = -[\bar{\delta}/(r+\bar{\delta}) + \bar{\delta}/(1-\bar{\delta})]\tilde{\delta}_0$. The flow value of a vacancy, and hence the cost of a hire, rises by $\bar{\delta}/(r+\bar{\delta})$ percent per percent of δ , which is 0.45 at D1. The value of a hire falls only through survival, by the small amount $\bar{\delta}/(1-\bar{\delta})$. For the two to stay equal, the expected time to fill, $1/q$, must fall, so tightness falls.

The iid case. Tightness and vacancies fall only at date 0, while unemployment is predetermined. At date 1 the flow value is back to $\bar{K}$ and $\theta_1 = \bar{\theta}$. Unemployment rises, and $v_1 = \bar{\theta}u_1$ rises with it as entry refills the stock. In the (u, v) plane the economy drops straight down, jumps up and to the right, and returns along the ray $v = \bar{\theta}u$. At D1, of the 0.16% rise in $\hat{u}_1$ per 1% shock, 0.04 comes from destruction, $(\bar{f}\bar{\delta}/\bar{\tau})\tilde{\delta}_0$, and 0.12 from the date-0 collapse in hiring, $\omega|\hat{\theta}_0|$.

The persistent case. Tightness stays low while destruction stays high, so vacancies stay below steady state while unemployment builds up. Once tightness has recovered enough, $v = \theta u$ turns positive and stays positive.

Where $\bar{\delta}$ relative to $\bar{\tau}$ enters. For the destruction shock, $\bar{\delta}$ enters both conditions only through R_δ , the ratio of the unemployment loading, $\bar{f}\bar{\delta}/\bar{\tau}$, to the tightness loading, $|a|$. The second is roughly proportional to $\bar{\delta}/(r + \bar{\delta})$, so R_δ is roughly proportional to $(r + \bar{\delta})/\bar{\tau}$. Lowering $\bar{\delta}$ relative to $\bar{\tau}$ shrinks the unemployment loading in proportion, while the tightness loading shrinks much less (Table 1), so R_δ falls. At D1, $(r + \bar{\delta})/\bar{\tau} = 0.19$ and $R_\delta = 0.045$, so the threshold $\omega + R_\delta$ is barely above $\omega = 0.164$. With $\bar{\delta} = \bar{\tau}$ (CK, $s = 0$), $(r + \bar{\delta})/\bar{\tau} = 1.11$ and $R_\delta = \bar{f}/|a| = 0.289$: the vacancy dies with the match, so its flow value cannot respond much more than unemployment does, and the threshold rises to 0.453.

The destruction–separation asymmetry. The two shocks share the denominator D of their tightness coefficients and the thresholds of Proposition 1. They differ in R_x . A destruction shock depreciates the vacancy asset, so tightness moves through the flow-value term $\bar{\delta}/(r + \bar{\delta})$, and at D1 its unemployment loading is small because destruction is a small share of separations. A separation shock leaves the vacancy’s flow value unchanged and moves tightness only through the probability that a match continues, while its unemployment loading is large because separations are most of the flows. At D1 and $\rho = 0.592$, $R_\delta = 0.045$ against $R_s = 9.56$. Both persistence terms are below one, so the separation shock gives positive comovement at every persistence, while the destruction shock gives negative comovement once it is moderately persistent. This is the free-entry counterpart of `prop:ds_asymmetry`, with closed-form conditions on both sides.

7. Numbers

The calibration uses $\rho = 0.592$, and responses are per 1% shock ($\tilde{\delta}_0 = 0.01$), in percent. At D1, $a = -0.790$ (iid: -0.759). At $\bar{\delta} = \bar{\tau}$, $a = -1.464$ (iid: -1.561).

Condition (21), $\rho > \omega + R_\delta$, holds in both economies, so the sign at $h = 1$ does not separate them. The path condition (23) does: 0.306 exceeds the threshold at D1 (0.209) but not at $\bar{\delta} = \bar{\tau}$ (0.453).

For the separation shock at D1 (there is none at $\bar{\delta} = \bar{\tau}$, where $\bar{s} = 0$), the unemployment loading is $\ell_s = 0.374$ and the bound (24) equals 4.05.

t	D1			CK-like ($\bar{\delta} = \bar{\tau}$)		
	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$
0	-0.79	0.00	-0.79	-1.46	0.00	-1.46
1	-0.47	0.17	-0.30	-0.87	0.66	-0.20
2	-0.28	0.19	-0.09	-0.51	0.76	0.25
3	-0.16	0.16	0.00	-0.30	0.66	0.36
4	-0.10	0.13	0.03	-0.18	0.51	0.33

Table 2: Impulse responses to a persistent destruction shock ($\rho = 0.592$).

	ω	R_δ	$\omega + R_\delta$	$\rho(1 - \lambda^2)/(1 + \rho\lambda)$	slope	ρ needed for slope < 0
D1	0.164	0.045	0.209	0.306	-0.24	$\rho > 0.37$
CK-like	0.164	0.289	0.453	0.306	0.17	none

Table 3: Conditions of Proposition 1 for the destruction shock at $\rho = 0.592$. The slope after an iid shock is $\bar{\theta} = 0.51$ in both economies.

A separation shock barely moves tightness, so vacancies rise almost one for one with unemployment from the first month on, the pattern of Proposition 1, part 3.

8. Remarks

Remark 2 (Holding K fixed). *If the vacancy cost were a constant flow, rather than the flow value of an asset that δ_t depreciates, the $\bar{\delta}/(r + \bar{\delta})$ terms would drop out of (14) and (20). Only the survival effect would remain, and $|a| = (1 + r)[\bar{\delta}/(1 - \bar{\delta})]/\{\eta_L[1 + r - \rho(1 - \bar{\tau})] + \rho\phi(1 - \bar{\delta})\bar{f}\}$ is small. At our calibration the path slope is then positive in every economy (about 0.49 at $\rho = 0.592$). This is the flow-cost DMP case. The separation shock never moves the flow value, so for it the durable vacancy and the flow-cost vacancy give the same tightness response.*

Remark 3. *A permanent shock ($\rho \rightarrow 1$) recovers the steady-state comparison. It hides the last statement of Proposition 1: without persistence, the response is positively sloped after impact.*

Remark 4. *prop:ds_asymmetry Part 3 in the draft gives “ $\bar{\delta}_e/\bar{\tau}$ small” as a sufficient condition. In the iid free-entry case, entry refills the stock at $t + 1$, and that condition fails for every $\bar{\delta}_e/\bar{\tau}$.*

ρ	b	R_s	$\omega + R_s$	$\rho(1 - \lambda^2)/(1 + \rho\lambda)$	slope
0.592	-0.039	9.56	9.73	0.306	0.50
0.874	-0.072	5.22	5.39	0.404	0.47

Table 4: Conditions of Proposition 1 for the separation shock at D1. The row $\rho = 0.874$ is the calibrated persistence of s .

t	$\hat{\theta}_t$	$\hat{u}_t$	$\hat{v}_t$
0	-0.07	0.00	-0.07
1	-0.06	0.39	0.32
2	-0.05	0.55	0.50
3	-0.05	0.60	0.56
4	-0.04	0.60	0.55

Table 5: Impulse responses to a separation shock at D1 ($\rho = 0.874$), per 1% shock, in percent.

Remark 5. *Part 3 of Proposition 1 is consistent with `prop:ds_asymmetry` Part 1 in the draft, and extends it. Part 1 signs the separation shock at $h = 1$ under conditions on entry, and solves the iid free-entry case exactly (its Step 4(a), where tightness stays at $\bar{\theta}$). Here the sign holds at every horizon and for every persistence, under (24).*